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Personalization, Engagement, and Content Quality on Social Media: An Evaluation of Reddit's News Feed

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Abstract. Digital platforms increasingly curate their content through personalized algorithmic feeds. Platforms have an incentive to promote content that increases the predicted engagement of each user to lift advertising revenues. This paper studies how ranking content to maximize engagement affects the credibility of news content with which users engage. In addition, I evaluate how the ranking algorithm itself can be designed to promote engagement with high-credibility content. Using data from the Reddit politics community, I exploit a novel discontinuity in the ranking algorithm to identify the causal effect of a post's rank on the number of comments it receives. I use this discontinuity to identify a model of user comment decisions and estimate the credibility of news content that users engage with under a personalized engagement-maximizing algorithm. The personalized engagement-maximizing algorithm exacerbates differences in the credibility of news content with which users engage. I then evaluate a credibility-aware algorithm that explicitly promotes credible news publishers and find the platform can substantially increase the share of engagement with high-credibility publishers for a small reduction in total engagement. These findings suggest algorithmic interventions can be a useful tool for managers to balance engagement quantity and content quality.

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Keywords: social media • digital platforms • ranking algorithms

1. Introduction

Social media platforms curate content for their users because of limited user attention and the vast amount of available content. The advertising business model adopted by many platforms creates an incentive to promote content through ranking algorithms that predict what content users are most likely to act on via clicking, liking, or commenting (Thorburn et al. 2022, Narayanan 2023). There is an active debate surrounding the benefits and potential risks of personalized rankings that optimize for engagement. Platform managers often contend that ranking algorithms act as agents for users by promoting a user's preferred content and reducing search frictions on the platform (Dorsey 2022). Critics, however, raise concerns that optimizing for engagement can incentivize low-quality or problematic content and reduce the diversity of viewpoints to which users are exposed (Pariser 2011, Orlowski 2020). These concerns have led policy makers around the world to consider regulating ranking algorithms (e.g., European Union's Digital Services Act). In addition, promoting low-quality content can be harmful to platforms if

advertisers respond by reducing advertising spending due to brand safety reasons (Ahmad et al. 2024) or if there is a disconnect between short-term engagement metrics and long-term user welfare (Spence and Owen 1977, Allcott et al. 2022, Kleinberg et al. 2023). Therefore, managers must balance maximizing engagement with the quality of content they are promoting to satisfy both internal and external stakeholders.

Despite these competing narratives, the impact of personalized news feeds on the quality of content users engage with remains an important and largely unresolved question. The lack of evidence regarding these issues primarily stems from the substantial challenges to studying ranking algorithms on social media platforms, including a hesitance to share data and experiments with external researchers and the identification challenges of observational approaches (Eckles 2022).

This paper analyzes how ranking algorithms on social media platforms impact the quality of news content with which users engage. To do so, I first estimate the causal effect of a news article's position in the feed on future engagement using a novel regression

discontinuity. The treatment effects demonstrate the importance of the ranking algorithm in steering attention and engagement to promoted articles. Second, I estimate a model of user engagement where the ranking algorithm influences engagement through attention as I hypothesize that users are less likely to be exposed to articles ranked further down the page. The estimates of the effect of rank on engagement identify how article rank impacts the likelihood of a user being exposed to a post in this model. Conditional on being exposed to an article, users choose whether to comment on the article depending on their preferences over article features. Users have heterogeneous preferences over article features and these heterogeneous preferences can be used by the platform to personalize content.

I then use this model to estimate engagement patterns under counterfactual ranking algorithms. First, I analyze how a personalized ranking algorithm that optimizes for engagement impacts the quality of content with which users engage. Second, I evaluate the cost to the platform in terms of foregone engagement of credibility-aware ranking algorithms that trade off optimizing for total engagement and engagement with high-quality content. Varying the weight placed on content quality allows me to trace the efficient frontier between engagement and content quality.

I explore this question in the context of political news on Reddit. In particular, I focus on the platform's largest politics community that centers on sharing and discussing news articles about U.S. political news. In this community, users share news articles about U.S. politics and then engage in discussion in comment threads alongside each article. I use the number of comments an article receives as a measure of engagement.

Reddit is an important platform to investigate the tradeoff between engagement and content quality for a number of reasons. First, Reddit is among the 10 most popular social media platforms in the world with more than 70 million daily active users (SimilarWeb 2024). In addition, Reddit is a critical component of the digital ecosystem playing an influential role in both web search and a source of training data for large language models (Patel 2024). As a result, Reddit is an important setting to understand how algorithms influence the content with which users engage.

The Reddit politics community also provides an ideal laboratory to analyze the content promoted under alternative ranking algorithms. The community is important to the platform due to its size.¹ In addition, it is often challenging to evaluate the quality of content on social media and a benefit of analyzing the politics community, which focuses on discussing news articles, is that I am able to use established measures of publisher credibility as a neutral measure of quality (Lin et al. 2022). Given this analysis is done independently without platform collaboration, I focus on this

particular measure of content credibility that is restricted to news publishers. That said, many platforms have internal measures of content and creator trustworthiness or credibility. For example, Meta has internal classifiers for untrustworthy and uncivil content (Guess et al. 2023). Therefore, the results and the algorithms introduced are relevant beyond this specific research setting. Finally, many key platform stakeholders, including advertisers, employees, and policy makers, have demonstrated an interest in the credibility of news content that appears on platforms that motivates the focus on a community where users share and discuss news articles.

Throughout the analysis, a central challenge will be the endogeneity of an article's rank: its position in the feed. One should be concerned that an article's potential outcomes are correlated with its position in the feed, because I expect the existing feed to promote articles that are more "commentable" relative to articles that are not promoted. Therefore, to identify position effects—the causal effect an article's position has on the number of comments it receives—I exploit a novel regression discontinuity revealed in an open-source mirror of the platform's code base that was maintained until 2017.² This open-source mirror allows me to inspect the ranking algorithm and recreate the numerical score that is used to rank articles. Consequently, this permits use of a regression discontinuity design to identify the local average treatment effect of an article's rank on the number of comments it receives in the subsequent period. As the ranking score of a focal article passes the score of a competing article, there is a discontinuous jump in the probability the focal article is ranked above the competing article on the page.³ The treatment effect estimates suggest that the causal effect of an article being promoted from the second position in the feed to the first position results in a 58.1% increase in the number of comments the article receives in the subsequent period. The effect of being promoted declines further down the feed.

With an identification strategy for position effects, I turn to understanding the impact of optimizing for engagement via personalized rankings on the quality of content to which users are exposed and with which they engage. To do so, I estimate a microfounded model of user comment decisions. I model engagement decisions based on two components: whether a user is exposed to an article and whether, conditional on exposure, their utility from commenting exceeds the utility of the outside option. Article rank impacts engagement in this model only through the exposure component, where the probability of being exposed to an article depends on the article's rank. Conditional on exposure, users then have heterogeneous preferences to comment on articles depending on the political slant and credibility rating of the publisher. The model allows for

arbitrary vertical preferences over article characteristics. This model is identified using the reduced form position effect estimates and individual engagement choices. I use the model to estimate engagement patterns under counterfactual ranking algorithms including both personalized and nonpersonalized engagement maximization. In addition, I evaluate alternative credibility-aware ranking algorithms that optimize for an objective function that balances total engagement and engagement with high-credibility publishers.

I find that a personalized engagement-maximizing algorithm exacerbates inequality in the quality of user news diets—the share of a user’s engagement with high-credibility publishers. The engagement maximizing algorithm tends to promote high-credibility publishers to users engaging with high-credibility publishers under Reddit’s actual ranking algorithm and promotes lower-credibility publishers to users engaging with less-credible publishers under the actual ranking algorithm. Moreover, I find that personalized engagement maximization leads to engagement with publishers that are less politically diverse and more similar to publishers the user has engaged with previously.

I next analyze engagement patterns under credibility-aware counterfactual ranking algorithms that optimize alternative objective functions that explicitly tradeoff total engagement and engagement with high-credibility publishers. At one extreme, this nests a credibility-maximizing algorithm that maximizes engagement with high-credibility publishers. This algorithm leads to a 5.4% decline in total user engagement. However, platforms can achieve over half of the increase in news diet quality from the credibility-maximizing algorithm for a more modest 1.9% decrease in engagement. This change in engagement is similar in magnitude to the difference between the personalized and nonpersonalized engagement-maximizing algorithms. However, the nonpersonalized algorithm does not meaningfully improve the quality of user news diets, whereas the credibility-aware algorithm increases the average user’s share of engagement with high-credibility publishers by 7.5 percentage points. This result suggests there is room for managers to use personalized ranking algorithms to balance the competing objectives of maximizing total engagement while maintaining the quality of content on the platform.

1.1. Related Literature

This paper primarily contributes to two strands of literature. A large literature has studied the impact of algorithmic recommendations on consumers. This literature considers algorithmic recommendations’ impact on product sales (Fleder and Hosanagar 2009, Oestreicher-Singer and Sundararajan 2012, Ghose et al. 2014, Hosanagar et al. 2014, Lee and Hosanagar 2019, Donnelly et al. 2023, Wang et al. 2025), content consumption (Holtz et al. 2020, Aridor et al. 2022, Chen

et al. 2023, Peukert et al. 2023), and consumer welfare (Ghose et al. 2014, Donnelly et al. 2023, Kaye 2024). In addition, this literature investigates how ranked feeds on social media platforms impact the emotional content of posts (Kramer et al. 2014), media consumption (Bakshy et al. 2015, Levy 2021, Dujeancourt and Garz 2023, Guess et al. 2023), and exposure to content from politicians (Huszár et al. 2022). Much of this literature explores the impact of algorithmic ranking on the diversity of consumption (Van Alstyne and Brynjolfsson 2005, Fleder and Hosanagar 2009, Holtz et al. 2020, Chen et al. 2023, Peukert et al. 2023) or product sales (Oestreicher-Singer and Sundararajan 2012, Hosanagar et al. 2014, Lee and Hosanagar 2019) and how likely users are to be exposed to cross-cutting news publishers (Bakshy et al. 2015, Levy 2021).

Most related, Huszár et al. (2022) analyzes an experiment on Twitter and Guess et al. (2023) analyze an experiment on Facebook and Instagram that randomly assigns users to receive a reverse chronological ranking algorithm relative to the existing personalized algorithm. Huszár et al. (2022) find that Twitter’s personalized algorithm amplified right-leaning publishers, and Guess et al. (2023) find that the reverse chronological feed increases exposure to political and content from untrustworthy sources. In addition, Beknazar-Yuzbashev et al. (2025) document in a field experiment a tradeoff between engagement and content toxicity on social media through a field experiment. This paper contributes to the literature by investigating the engagement-credibility tradeoff the platform faces. In addition, the approach I take allows for more granular comparisons between algorithms allowing me to isolate the impact of personalization and the algorithm objective on the type of content with which users engage. An additional benefit of the modeling-based approach taken here is that it allows me to consider many counterfactual ranking algorithms without the costs of testing each ranking algorithm experimentally. A final contribution to this literature is that I study the impact of ranking algorithms on the credibility of content with which individuals engage independent of platform collaboration, which involves unique challenges around research design and data collection.

Second, this paper contributes to the large and growing literature studying interventions to improve the quality of information people consume online. This literature both documents the reach of misinformation on social media and how it spreads (Allcott and Gentzkow 2017; Vosoughi et al. 2018; Grinberg et al. 2019; Guess et al. 2019, 2020) and evaluates interventions to curb the spread of misinformation (see Pennycook and Rand (2021) and Lazer et al. (2018) for a review). My findings are consistent with the literature showing that a minority of users account for a large share of consumption of low-credibility news content (Allcott and Gentzkow

2017; Grinberg et al. 2019; Guess et al. 2019, 2020) and contribute to the literature by finding that personalized engagement-maximizing algorithms exacerbate this difference. In addition, this literature assesses many behavioral interventions through both laboratory and field experiments. That said, empirical evaluations of algorithmic interventions have been more difficult given limited access to platform data. I contribute to this literature by exploring the impact of algorithmic interventions that promote high-credibility publishers and estimate the cost of such interventions.

2. Background and Data

Reddit is a large social news aggregator with more than 70 million daily active users as of January 2024 and was valued at approximately \$6.5 billion shortly after its initial public offering in 2024.⁴ The platform is organized into more than 100,000 virtual communities called subreddits that are focused on sharing and discussing content related to the community's topic. In this study, I focus on the politics community that is a community that is centered around sharing and discussing news articles about U.S. politics. In this community, users share news articles and then discuss the articles in comment threads. Reddit is structured such that users can submit two types of content: submissions and comments. In the politics community, submissions must contain a link to a news article, and I therefore use the terms submissions, articles, and posts interchangeably. Users then discuss articles by posting comments, and this commenting activity is the primary engagement measure I study. Given that users who post an article are unable to add any discussion or commentary on their submission, I focus on analyzing features of the article's publisher rather than the user who submitted the article. I therefore refer to the publisher of an article as the news outlet that published the story rather than the user who submits the article.

2.1. Algorithmic Feeds on Reddit

Users interact with content on the platform via algorithmic feeds of a few different forms. Any user who visits a community page will see submissions from the community ranked by the platform's default ranking algorithm.⁵ This algorithm sorts submissions according to a ranking score that combines the post's age and vote score—the net number of upvotes minus downvotes on a post—and is described in more detail in Section 3. In addition to the default algorithm, users can choose to rank posts according to several alternative algorithms. The *new* algorithm implements a reverse chronological ranking; the *top* algorithm ranks posts according to the vote score in a given period; the *rising* algorithm favors recent posts; and the *controversial*

algorithm promotes posts that have received more votes, either up or down, regardless of their direction. This paper focuses on the default algorithm's impact on engagement. All analyses presented here are conditioned on the alternative algorithm rankings remaining unchanged. That is, when estimating the impact of post rank on engagement, I estimate counterfactuals where post rank changes in the default feed but not in the alternative feeds.

I estimate in a supplementary survey that over half of users in the Politics community primarily use the default feed (Figure A.5 in the Online Appendix). The remaining users who do not use the default feed primarily use the *new* algorithm. These users are unlikely to influence my analysis for two reasons. First, these users are unlikely to be in my sample. The median post in my sample is 9.8 hours old. Given the volume of posts on the politics community, there are 301 newer posts than the median post in my sample. This implies the median post in my sample is on the 12th page of the *new* algorithm. It is unlikely that these posts are driving much engagement and therefore these users are unlikely to reach the engagement threshold, which I describe in Section 4, to be included in my analysis. Even the fifth percentile post is 1.8 hours old and has 51 newer posts preceding it in the *new* algorithm. Second, even if a user was included in the data set, it is unlikely that this algorithm is influencing the treatment effects I estimate given how far down the *new* algorithm the posts I consider are ranked. As I will show, the effect of rank on engagement quickly dissipates lower down the feed so it is likely the causal effect of rank on engagement is negligible beyond the first page of posts.

Comments on articles are also algorithmically ranked on Reddit using similar ranking algorithms. The rank of articles, however, does not directly depend on the number of comments the article has or the number of votes that comments have received. Only votes on the articles and the age of the article impacts ranking of articles as I describe in detail in Section 3. I focus on the ranking of articles throughout this analysis and treat the ranking of comments as fixed.

Reddit users can also join communities. Posts from these communities are displayed on a user's Home feed, the default feed users encounter when visiting the platform. The Home feed sorts posts according to the same default algorithm used by individual communities but ranks posts from all communities that a user is a member of rather than only posts from a single community.⁶

The structure of the Home feed has important implications for this analysis. The empirical strategy estimates the effect a post's rank in the subreddit feed has on its future engagement and how alternative ranking algorithms on the subreddit feed impact engagement

patterns. Changing the rank of a post impacts engagement with that post through two channels. First, there is the direct channel on users who interact with posts through the subreddit feed. Second, changing the rank of a post in the subreddit feed also changes the order of the post in the home feed holding fixed the position of posts from nonaffected communities. In other words, the treatment effects estimate the effect of rank on engagement in both the subreddit feed and in the home feed. Moreover, when I analyze engagement patterns under counterfactual ranking algorithms, the rank of posts are being changed in both the subreddit feed and the Home feed.

To make this concrete, consider a simple example where there are three posts on the platform. Post *A* and post *B* are from the politics community, and post *C* is from a different community. On the Home feed the posts are ranked *A, C, B*, which also implies that post *A* is ranked above post *B* in the politics community. In a counterfactual where post *B* is promoted ahead of post *A* in the politics community, this also corresponds to a change in the Home feed ranking to *B, C, A* given the two feeds use the same algorithm to rank posts. Crucially, the position of the nonpolitics community post *C* is unchanged in this counterfactual because the counterfactuals only vary the relative ranking of the politics posts *A* and *B*, holding fixed the ranking scores, and hence positions, of all nonpolitics posts. This interpretation follows directly from the regression discontinuity identification strategy described below, which estimates the causal effect of rank on engagement using quasi-random variation in whether post *A* or post *B* is ranked first on the politics feed holding all other factors fixed. In practice, post *C*'s vote trajectory and time decay would continue to evolve over time, but the counterfactuals compare outcomes under two different politics rankings at the same point in time, with post *C*'s characteristics held fixed. Given the prominence of the Home feed, it is important that the position effect estimates and counterfactual analyses include the effect of post rank in the Home feed.⁷

2.2. Data

2.2.1. Ranking and Engagement. I merge data from several sources in this study. First, I scrape subreddit landing pages from the Internet Archive's Wayback Machine for each subreddit in the study.⁸ These data provide historical snapshots of subreddit feeds, allowing me to collect the top 25 ranked posts, their position in the feed, and post features. Note the Wayback Machine only takes a snapshot of the first page of the subreddit feed. Therefore, I restrict to the top 25 ranked posts in my analysis. I discuss the implications of this decision in Section 5.1. A snapshot from the politics community following the 2016 election is shown in

Figure 1. The large blue box contains the main feed, which consists of a list of posts. Alongside the post position in the feed, parsing the Wayback Machine snapshots provides the age of a post, number of existing comments, the vote score of each post (net number of upvotes minus downvotes, shown in the orange box), post title, and domain the post links to, if any. In addition, each snapshot reveals the number of subscribers each community has and the number of users online at the time of the snapshot (highlighted in green box at the bottom right of the image). The Wayback Machine snapshots are unevenly distributed over time (Figure A.1 in the Online Appendix). The exact frequency of snapshots depends on a number of factors, however, comparing the frequency of snapshots with the number of comments posts receive in the 60 minutes following each snapshot shows a correlation between snapshot frequency and the number of comments. Therefore, I interpret the snapshots as being roughly proportional to the traffic the platform receives. Approximately half of snapshots occur in 2016, with another 35% occurring earlier in 2013. Therefore, the data can be interpreted as being most reflective of users active on the platform in these two years. Figure A.2 in the Online Appendix plots the share of articles that appear in multiple snapshots. Approximately half of articles appear in a single snapshot, and more than 75% of articles appear in at most two snapshots.

Submissions on Reddit are either pinned to the top of the feed by community moderators or ranked organically.⁹ I focus on organic posts displayed in blue in Figure 1. These posts are submitted by users and ranked according to the algorithms described in Section 2.1. In the politics community considered here, posts are required to follow strict community guidelines: They must be on topic for the community, they must link to an article from a news publisher, and the post title must exactly match the headline of the article to which the post links. Any commentary on the article must be added in the comment sections, which I turn to next.

The primary engagement metric I consider is comments on articles. I do this for both practical reasons—because data on comments are readily available, whereas high-quality data on votes or views are not publicly available—and for substantive reasons that I detail below. In particular, Reddit is a platform centered around sharing and discussing user-generated content. Comments themselves are a form of user-generated content that bring people to the platform, and encouraging additional comments is of direct interest to the platform (Burke et al. 2009). In addition, experimental evidence suggests users who receive comments on their posts are more likely to generate content in the future, a finding that further supports the premise that encouraging more comments is desirable for Reddit (Eckles et al. 2016, Mummalaneni et al. 2026).

Figure 1. (Color online) Snapshot of Politics Community

Notes. A Wayback Machine snapshot of the politics subreddit from November 2016. The large blue box contains the main feed. Posts pinned by moderators are shown in green and are typically threads created to discuss major events or frequent discussion topics such as polling. Posts in blue are algorithmically ranked organic posts that are the focus of this study. The orange box on the left highlights the vote score of each post. The small green box in the bottom right highlights the number of subscribers to the subreddit and the number active at the time of the snapshot.

In addition, there is substantial evidence that commenting is highly correlated with other forms of engagement including taking actions such as voting or liking content, viewing a piece of content, and the amount of time spent on a platform. This includes a strong correlation between voting and comment behavior on Reddit that is described in Section A.1 in the Online Appendix. The literature also provides substantial evidence of a high correlation between engagement measures (Wojcik and Hughes 2019), including experimental evidence from Facebook and Instagram showing that treatment effects from changes to the news feed algorithm are qualitatively similar across many forms of engagement including time spent, viewing, liking, and commenting (Guess et al. 2023).

I merge data from Baumgartner et al. (2020) that contain a near-universe of submissions and comments to public Reddit communities to generate the engagement

outcomes. These data contain user-level commenting behavior, where each comment includes a timestamp, a user identifier of the comment author, the full text of the comment, the post the comment is responding to, and the vote-score the comment received, among other observables. This information allows me to reconstruct a post's full comment history, including the comments that occurred immediately following each of the Wayback Machine's snapshots.

These data on user comments serve several purposes. First, they allow me to construct the number of comments each post received in a window following each snapshot. This will be critical for estimating effects of position on engagement on the platform. Second, individual-level comment decisions are used to estimate a choice model of user engagement in Section 4. Here, the panel nature of the data allows me to identify rich user-level heterogeneity in comment preferences.

Finally, studying comments allows me to analyze the text content to provide additional insight into user preferences and to understand how optimizing for engagement impacts the sentiment of comments submitted to the platform (See Online Appendix D).

2.2.2. Publisher Slant and Credibility Ratings. I also collect two sets of publisher ratings that capture various aspects of an article's publisher. First are measures of a publisher's political slant (Robertson et al. 2018) that represent the relative propensity of a publisher domain being shared on Twitter by known Democratic party members relative to known Republican members, ranging from -1 to 1 . A slant rating of -1 represents a domain that is only shared by Democrats, whereas a slant rating of 1 represents a domain that is only shared by Republicans. Robertson et al. (2018) demonstrates this measure is consistent with a number of other expert, crowd-sourced, and audience-based ratings (Bakshy et al. 2015, Budak et al. 2016). A primary benefit of the Robertson et al. (2018) scores compared with other measures of publisher slant is the high coverage, because the data set includes ratings for more than 19,000 domains.

I use credibility ratings, described in Lin et al. (2022), for 11,520 news publishers. Lin et al. (2022) aggregate individual ratings from six rating organizations and demonstrate substantial agreement among individual sources. Importantly, the ratings released alongside Lin et al. (2022) show an extremely high correlation with NewsGuard ratings, a proprietary set of publisher ratings that employ extensive criteria including accuracy and balance of reporting, a process of publishing corrections, clear separation of opinion articles, and transparency of perceived conflicts. Figure A.3 in the Online Appendix plots the joint distribution of publisher slant score and credibility rating for publishers that appear in at least 1% of the snapshots in the politics community. Table A.1 in the Online Appendix shows these ratings for five example domains. In evaluating user news diets, I discretize the credibility ratings into high- and low-credibility publishers for ease of interpretation. When doing so, I classify publishers as high credibility if their credibility rating is greater than 0.65, and I show robustness of key results to other thresholds in Section C.4 of the Online Appendix.¹⁰

I successfully match 90% of articles with a credibility rating and 94% of articles with a slant score. I mean-impute the ratings for articles when the publisher cannot be matched.

2.3. Model-Free Evidence of an Engagement-Credibility Tradeoff

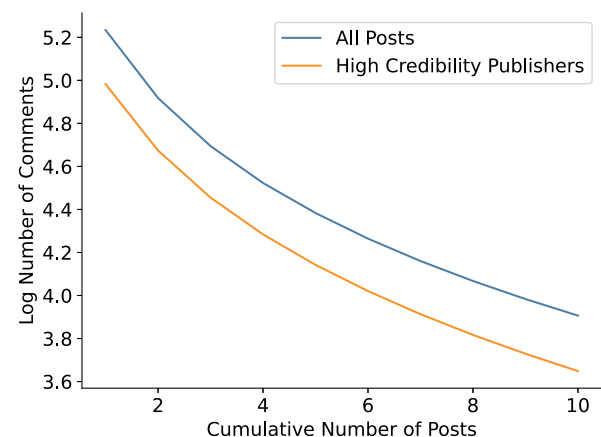
I next provide a model-free analysis to motivate further study of how engagement maximizing algorithms impact the credibility of publishers with which users

engage and the tradeoff platforms face between these quantities. Although articles by low-credibility publishers are roughly equally as engaging as articles by high-credibility publishers on average, a tension remains between credibility and engagement at the efficient frontier. To see this, Figure 2 plots the average log number of comments received by the top K most popular posts in each period in the hour following a snapshot. I plot this separately for all articles and articles by high-credibility publishers. There is a clear divergence in the two lines, suggesting that there are many periods where articles by low-credibility publishers receive substantially more comments than those by high-credibility publishers. An algorithm that optimizes for engagement would likely promote such articles above articles from high-credibility publishers, presenting a tradeoff between engagement and the credibility of content with which users engage. Section 5.2 quantifies the extent of the tradeoff between engagement and credibility using a structural model of user engagement decisions.

3. Effect of Rank on Engagement

In this section, I estimate the causal effect post rank has on the number of comments the post receives. Recall that I ultimately want to understand how engagement-maximizing ranking algorithms impact the type of content with which users engage and a key ingredient to this analysis is the causal effect of rank on engagement. Naive comparisons between posts ranked above a competing post with the post ranked below are unlikely to identify the causal effect of rank as I expect potential

Figure 2. (Color online) Cumulative Average Engagement



Notes. This figure plots model-free evidence of the trade-off between engagement and engagement with high-credibility publishers. Each line plots the average log number of comments by the top K most popular articles within the 60 minutes following a snapshot. The x -axis (K) varies from 1 to 10. The All Posts line plots the average of the top K most popular posts for all posts in the period. The High-Credibility Publishers line restricts to only posts by high-credibility publishers.

outcomes to be correlated with post rank (Narayanan and Kalyanam 2015, Ursu 2018). Specifically, it is likely that posts with high potential outcomes are more likely to garner upvotes as the posts may be more inherently interesting. As I will describe, upvotes play a central role in the ranking algorithm and thus these posts are likely to be shown higher on the page.

This section introduces the identification strategy I use to overcome this challenge by estimating position effects: the causal effect of post rank on engagement. This serves two purposes. First, the treatment effect estimates provide important motivation for the analysis, given that I find post rank has a large causal effect on engagement, meaning the ranking algorithm plays an important role in shaping the posts with which users eventually engage. Second, the causal estimates from this section will be utilized directly in identifying the choice model of user engagement that is employed to analyze counterfactual ranking algorithms.

I exploit a regression discontinuity to identify the causal effect of rank on engagement. Until 2017, Reddit maintained an open-sourced mirror of its code base, which allows me to directly inspect the algorithm used to sort posts (Reddit 2017). The algorithm assigns a ranking score to each post and ranks posts in descending order of these scores. Formally, a post's ranking score is defined as

$$s_{jt} = \text{sign}(s_{jt}^v) \log_{10}(\max\{|s_{jt}^v|, 1\}) - \frac{a_{jt}}{45,000}, \quad (1)$$

where s_{jt}^v is the net number of upvotes minus downvotes that post j had at time t and a_{jt} is the age of the post in seconds. This requires that for the ranking score of a post with a positive vote score to remain constant, every 12.5 hours the net number of upvotes minus downvotes must increase by a factor of 10 to offset the age penalty. Importantly, this defines a continuous score that determines post rank, creating a regression discontinuity that can be used to identify position effects. As the open-source mirror was only maintained until early 2017, I only use data through 2016 for all analyses.

To give a concrete example of the regression discontinuity I exploit, consider two adjacent posts i, j with ranking scores s_i, s_j and observed ranks r_i, r_j . There is a discontinuous jump in the probability of post i being ranked above post j when the continuous forcing variable $s_i - s_j$ crosses zero. I take advantage of this discontinuity to identify the effect of rank on future engagement, under the assumption that potential outcomes (i.e., latent post quality) are continuous across the zero threshold of the forcing variable ($s_i - s_j$).

3.1. Implementation Details of the Regression Discontinuity Design

I now discuss the implementation details of the regression discontinuity used to estimate the causal effect of

post rank on future engagement. In particular, I focus on estimating the causal effect of moving up from position $r + 1$ to position r on the feed. To simplify notation, let D_i be a treatment indicator (i.e., $D_i = 1[s_i > s_{-i}]$), and the forcing variable is denoted $\Delta s_i = s_i - s_{-i}$.

In this setting, the running variable is a composition of two scores: the ages and vote scores of the posts. This creates a cutoff frontier, shown in Figure B.13 in the Online Appendix, analogous to geographic regression discontinuity designs. I take advantage of the multiple score nature of the problem and estimate the treatment effect at the origin, which ensures that posts are balanced on both post age and vote score, as described in Cattaneo et al. (2024).

The primary results use a quasi-Poisson regression with a linear term for the running variable to model the conditional expectation functions on either side of the discontinuity and a uniform kernel. I will show the estimates are similar under alternative specifications. I restrict to observations within a bandwidth λ of the cutoff chosen to minimize the mean squared error of the treatment effect estimator (Calonico et al. 2014, Cattaneo et al. 2020) and demonstrate the results are not sensitive to this choice (Section B.2 in the Online Appendix).¹¹ I winsorize comments at the 95th percentile of comments for each adjacent pair of posts.

For each of the first 24 positions on the first page of the feed, I estimate the treatment effect of moving from position $r + 1$ to position r as

$$\hat{\tau}_r = \hat{\mu}_r^+ - \hat{\mu}_r^-, \quad (2)$$

where $\hat{\mu}_r^+$ is the estimated intercept from the quasi-Poisson regression to the right of the discontinuity and $\hat{\mu}_r^-$ is the intercept to the left of the discontinuity (Cattaneo et al. 2020).¹² I estimate the treatment effect separately for each position in the feed, allowing the treatment effect of being promoted from position $r + 1$ to position r to vary by position. Table B.2 in the Online Appendix assesses the sensitivity of the results to the degree of polynomial terms included in the regression function and to using a linear specification with the log of one plus the number of comments an article receives as the outcome variable. I cluster standard errors at the period level to allow for within period correlation.

The rank-specific estimates are imprecise, so I also estimate a single model that parameterizes the treatment effect curve to smooth the estimates. Specifically, I stack the data within the bandwidth for the 24 boundaries (indexed by r) and estimate the following Poisson regression:

$$\log E[Y_i | D_i, \Delta s_i, r_i] = \psi_{r_i} + \alpha_{r_i} \times \Delta s_i + \gamma_{r_i} \times \Delta s_i \times D_i + \tau_0 D_i + \tau_1 \log r_i \times D_i, \quad (3)$$

where Y_i is the number of comments article i received

in the 60 minutes following the snapshot, ψ_r are fixed effects for rank boundary r , α_r captures the slope of the regression function with respect to the running variable below boundary r , γ_r captures the change in the slope of the regression function with respect to the running variable above boundary r , τ_0 is the intercept for treatment effect curve and τ_1 is the slope of the treatment effect curve with respect to $\log r$. We can then estimate the smoothed local average treatment effect of moving from any position $r + 1$ to position r on the feed as $\tau_0 + \tau_1 \log r$.

3.1.1. Measurement Error in the Running Variable. A challenge in this setting is that the running variable is constructed using data scraped from the Internet Archive's WayBack Machine and the reconstructed ranking scores do not completely determine the rank of a post. This is a result of several factors. First, Reddit explicitly adds noise to the vote scores shown to users to combat vote manipulation (Muchnik et al. 2013).¹³ Second, Reddit caches votes and rankings for performance purposes due to the large amount of traffic the platform receives.¹⁴ Caching means the ranking score and actual ranks are not continuously updated. Finally, many digital platforms run experiments that can include ranking experiments. Combined, this makes it possible for the observed ranks to differ from what is implied by the relative ranking scores as either the scores or observed rankings are a cached version.

Adding noise to the vote score introduces measurement error into the running variable and can bias traditional regression discontinuity estimates. To estimate the local average treatment effect of rank in the presence of measurement error in the running variable I follow Dong and Kolesár (2023) by excluding posts within a doughnut around the discontinuity. I manually select a doughnut width of 0.05 on either side of the cutoff and show in Section B.2 in the Online Appendix that the results are robust to the choice of doughnut width. Under the assumption that the doughnut excludes all periods where the posts are misclassified due to measurement error, Dong and Kolesár (2023) show that the usual regression discontinuity estimators identify a local average treatment effect.

After excluding posts within a doughnut of the discontinuity, I assume the remaining mismatch between post rank and the relative ranking scores is not due to measurement error. Remaining mismatch is likely due to caching vote scores and positions for performance purposes and experimentation by Reddit. This assumption appears justified, as the probability of mismatch is constant as one moves away from the discontinuity (Figure 3(a)). If this was driven by the noise added to vote scores, the probability of mismatch would decline further away from the discontinuity because the probability the noise added is sufficiently large to misclassify the posts declines. Therefore, estimating the local

average treatment effects using local quasi-Poisson regression results in conservative estimates of position effects. This has little impact in practice, because the estimated first-stage effect is close to one in most cases.

3.2. First Stage and Balance of the Regression Discontinuity Design

I now show evidence that Reddit ranks posts according to the algorithm I describe to establish a first stage in the regression discontinuity analysis. For each position on the feed $r \in \{1, \dots, 24\}$, I consider the two posts ranked in position r and $r + 1$ and plot the probability that a post is in position r against the running variable (the difference in ranking scores of the two competing posts). Figure 3(a) shows the discontinuity between position 1 and position 2; the plots for the remaining positions are shown in Section B.1 in the Online Appendix. There is a clear discontinuity in the probability a focal post is ranked above a competing post when the focal post's ranking score surpasses that of the competing post.

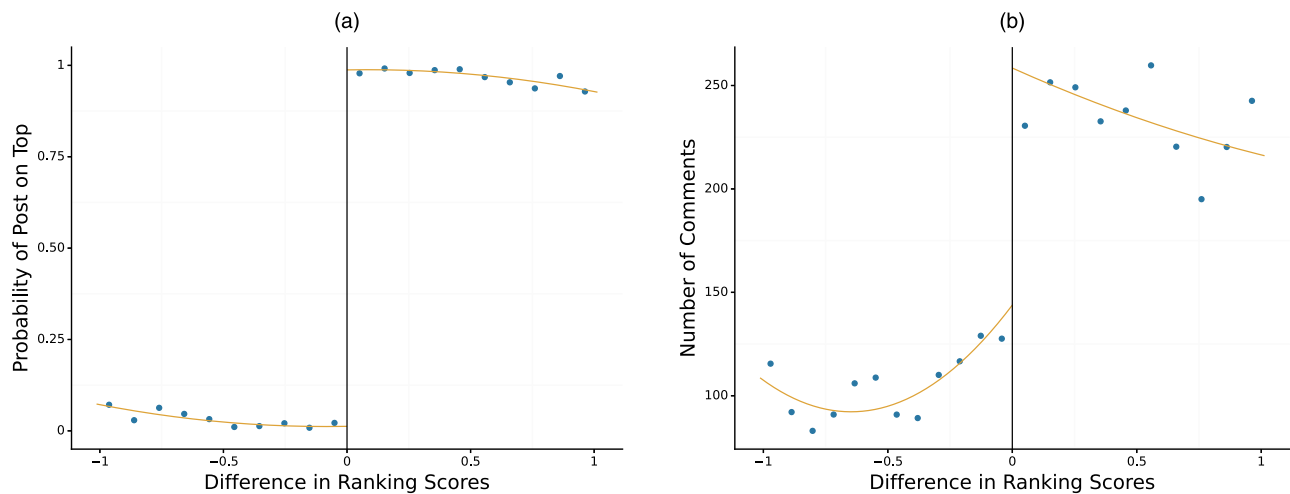
In addition, to test that post observables are balanced across the discontinuity, Figure B.18 in the Online Appendix plots the estimated treatment effect of rank on pretreatment covariates including publisher slant, publisher credibility, post vote score, and post age. Nearly all estimates are insignificant at the 5% level, suggesting post observables are balanced across the discontinuity. Although it is not possible to test the identifying assumption that potential outcomes are continuous through the discontinuity, this result is consistent with such an assumption holding. Section B.2.2 in the Online Appendix presents plots of the nonparametric conditional expectation functions of these covariates around the discontinuity.

3.3. Position Effect Estimates

I now turn to estimating how post rank affects the engagement a post receives in the window following the snapshot. Figure 3(b) plots a binned scatter plot of the average number of comments a post receives in the 60 minutes following each snapshot against the running variable (Δs_i) to visualize the discontinuity in the outcome variable. There is a clear discontinuity in engagement when a post is promoted to position 1 from position 2. Section B.1 in the Online Appendix shows the same plots for the remaining positions on the feed and the discontinuity in engagement quickly disappears further down the feed, suggesting treatment effects of rank are largest at the top of the feed.

I estimate the local average treatment effect using local quasi-Poisson regression and present treatment effect estimates in Figure 4. This figure plots the local average treatment effect of moving from rank $r + 1$ to rank r on the log number of comments a post receives in the 60 minutes following a snapshot. Being promoted to the first position has a large effect, with the

Figure 3. (Color online) Regression Discontinuity Plots



Notes. (a) Effect on post rank. (b) Effect on engagement.

treatment effect estimate suggesting a 58.1% increase in the number of comments received immediately following a snapshot relative to the second post. The importance of rank quickly dissipates further down the feed.

Figure 4 includes naive position effect estimate. As expected, the naive estimates overestimate the effect of position on engagement, although this overestimation is concentrated at the top of the feed.¹⁵ Outside of the top position on the feed, the naive estimates and the regression discontinuity based estimates are remarkably similar. This suggests that the endogeneity concern is most prevalent for the top article within each period, which is consistent with there being a top story within each period that is ranked at the top of the feed and that would have garnered substantial engagement irrespective of its position.

In addition, Figure 4 makes clear that rank-specific estimates are imprecise. The smoothed estimates from Equation (3) are also contained in Figure 4, and the low-dimensional parameterization closely fits the individual treatment effect estimates. Table B.4 in the Online Appendix contains estimates of τ_0 and τ_1 from Equation (3) and the smoothed τ_r treatment effect estimates that I use moving forward.

These treatment effects demonstrate that the ranking algorithm has an important effect in determining the posts with which users engage. This in turn motivates further investigation of how the design of ranking algorithms impact the credibility of content users engage with as the platform has substantial power in determining what content users are exposed to and ultimately engage.

4. Model of Individual Engagement Decisions

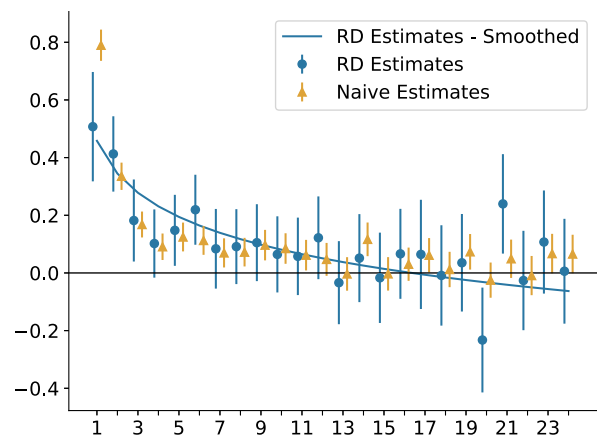
I now estimate a model of user engagement that allows me to estimate engagement patterns under counterfactual

ranking algorithms. Section 4.1 introduces the model of individual decisions, Section 4.2 describes identification and the estimation approach, and Section 4.3 summarizes the model estimates and fit.

4.1. Model

The model of user engagement decisions has several components. First, users visit the platform and are exposed to a subset of articles that is partially determined by the ranking algorithm. In this model, the ranking algorithm affects engagement by focusing user attention on articles promoted in the feed. Conditional on being exposed to an article, a user then comments if

Figure 4. (Color online) Position Effect Estimates



Notes. Estimates of the local average treatment effect from a post moving from position $r + 1$ to position r on the log of the expected number of comments a post receives in the 60 minutes following a snapshot. The points labeled RD are from the regression discontinuity estimates and the points labeled Naive are from a naive comparison of articles ranked higher to those ranked lower. Error bars represent 95% confidence intervals using robust standard errors clustered at the period level.

their utility from doing so exceeds the outside option. This model flexibly allows for users to have arbitrary vertical preferences and heterogeneous preferences over article features. This heterogeneity is important, as many of the counterfactual ranking algorithms I evaluate take advantage of personalization where preference heterogeneity may result in different rankings for different users.

Formally, users indexed by i visit the platform in periods indexed by t and are exposed to a ranked feed of posts indexed by j . In each period, users are exposed to a post in position r_{jt} if $v_{ijt} = 1$, which is an independent Bernoulli random variable equal to one with probability $p(r, t)$. I use a parsimonious parameterization of the exposure probability, $p(r, t) = p_t p_r$, where p_t is the probability of accessing the platform in period t and p_r is the probability of being exposed to a post in position r conditional on accessing the platform.¹⁶ If exposed to a post, users receive utility

$$u_{ijt} = \delta_{ijt} + \varepsilon_{ijt} \quad (4)$$

if they comment on post j in period t , which I denote $d_{ijt} \in \{0, 1\}$. Users comment if they are exposed to a post and the utility from commenting exceeds the utility of the outside option (u_{i0t}), which can be modeled formally as¹⁷

$$d_{ijt} = 1[v_{ijt} = 1]1[u_{ijt} \geq u_{i0t}]. \quad (5)$$

I model $\delta_{ijt} = \mathbf{x}'_{jt}(\bar{\boldsymbol{\beta}} + \boldsymbol{\beta}_i) + \xi_{jt} = \delta_{jt} + \mathbf{x}'_{jt}\boldsymbol{\beta}_i$, where $\mathbf{x}_{jt}$ is a vector of observable article features, $\bar{\boldsymbol{\beta}}$ represents average preferences, and $\boldsymbol{\beta}_i$ is a vector of the deviation of user i 's preferences from the mean.¹⁸ Finally, ξ_{jt} is an article-period fixed effect that represents latent article commentability. This fixed effect flexibly captures anything users have vertical preferences over.¹⁹ For example, the model allows all users to prefer to comment on articles with many existing comments or votes or to prefer articles on certain topics, all these preferences would be captured by the ξ_{jt} term. Table C.7 in the Online Appendix regresses the ξ_{jt} estimates onto additional article features to demonstrate how this term flexibly allows for arbitrary vertical preferences. Moreover, ξ_{jt} is article-period specific that flexibly allows for the commentability of an article to vary across periods (Figure C.26 in the Online Appendix). Post rank is excluded from utility in this model, implying that rank does not impact choice conditional on exposure to a post.²⁰ Finally, normalize $\delta_{i0t} = 0$ without loss and assume ε_{ijt} is an independent and identically distributed type 1 extreme value preference shock. This results in the mixed logit choice probabilities multiplied by the exposure parameter $p(\cdot)$:

$$P_{ijt} = P(v_{ijt} = 1, u_{ijt} \geq u_{i0t}) = p(r_{jt}, t) \frac{\exp \delta_{ijt}}{1 + \exp \delta_{ijt}}. \quad (6)$$

In the main text, I allow users to have heterogeneous preferences over publisher slant, publisher slant

squared, and publisher credibility. In addition, I allow for a heterogeneous constant to capture individuals' varying propensity to comment on articles. The set of features in which I allow heterogeneous preferences is intended to capture the key elements of heterogeneity in which I am interested. In principle, one could allow for arbitrary individual preference heterogeneity over publishers by allowing for an individual-publisher fixed effect though the demands on the data of such a model are severe. Allowing for heterogeneous preferences over additional post features (such as the vote score, post age, and the number of existing comments) does not materially change the findings (Section C.4.1 in the Online Appendix). To reiterate, however, the primary specification does allow for vertical preferences over arbitrary post and publisher features given I include a postperiod fixed effect (ξ_{jt}). This implies that user utility from commenting (and hence choices) can depend on arbitrary article and publisher features, including the number of existing comments and votes for example. This is shown in Table C.7 in the Online Appendix, which projects ξ_{jt} on additional post observables. Figure C.28 in the Online Appendix plots the distribution of the average ξ_{jt} by publisher.

This model captures key elements of user engagement decisions that are relevant for the counterfactual ranking algorithms in a tractable manner. First, the model allows for ranking algorithms to impact engagement decisions by steering attention to promoted articles. Second, the model allows for rich heterogeneity in preferences over publisher features without distributional assumptions as are often required in mixed-logit models (Berry et al. 1995). In particular, the panel data allow me to estimate a separate $\boldsymbol{\beta}_i$ for each user. Given a leading counterfactual will consider personalized ranking algorithms, it is important to capture such preference heterogeneity in the model and allow for the platform to personalize rankings based off these preferences.

Finally, the main text considers the binary decision of whether a user posts a comment on an article. This is to maintain focus on the primary research question of how ranking algorithms impact engagement with articles of varying publisher credibility. Online Appendix D considers a model where, in addition to choosing whether to comment, users also choose the sentiment of their comments they post. I also analyze how the various counterfactual ranking algorithms impact the sentiment of engagement.

4.2. Identification and Estimation

4.2.1. Identification of Model Parameters. The key identification challenge in this model is that observed post ranks are correlated with latent commentability, or $E[\xi_{jt}r_{jt}] \neq 0$. I now describe how this model is

identified using the regression discontinuity from Section 3 and user-level engagement decisions.

I first describe how the exposure parameters (p_t , p_r) are identified. I assume that each user logs on to the platform with probability p_t , independent of article features or preference shocks. This probability is identified via the share of users who visit the platform in each period that I estimate using data on the share of users online at the start of each period. Conditional on accessing the platform, I assume all users are exposed to the top post on the feed implying that $p_1 = 1$.²¹ Under the model's restrictions, the remaining exposure parameters p_r are identified by the reduced form treatment effects assuming constant treatment effects of rank on engagement. That is, I assume $E[Y_{jt}(r)] = e^{\tau_r} E[Y_{jt}(r+1)]$, where $Y_{jt}(r) = \sum_i d_{ijt}(r)$ is the potential outcome under rank r for the total number of comments post j in period t receives following a snapshot. This implies the following mapping between treatment effects and the ratio of exposure probabilities p_r :

$$\tau_r = \log \frac{\sum_i E[d_{ijt}(r)]}{\sum_i E[d_{ijt}(r+1)]} = \log \frac{p_r}{p_{r+1}}. \quad (7)$$

Importantly, Equation (7) relies on the assumption that post rank is excluded from utility, which is consistent with the existing literature on consumer search as discussed in Endnote 20. Under this assumption, the only reason an article's engagement would differ as its rank varies is due to the impact rank has on exposure. This allows the model to infer how article rank impacts exposure from the treatment effects of rank on engagement.

The assumption of constant treatment effects could in principle be relaxed to allow for arbitrary individual heterogeneity and heterogeneity along observed article features, although the demands on the data grow substantially if this type of heterogeneity are included. For example, allowing for arbitrary individual heterogeneity would require estimating the reduced form treatment effects separately for each user.

Given exposure parameters, individual- and mean-preference parameters are identified from the assumption that article features are exogenous $E[x_{jt}\varepsilon_{ijt}] = 0$ and $E[x_{jt}\xi_{jt}] = 0$. Importantly, the latent commentability of a post is not identified without the exposure parameters. Intuitively, without knowing the probability a user is exposed to a post one cannot attribute a low comment rate on a post to low commentability or low exposure. Therefore, the exposure probabilities are required for identifying the model.

4.2.2. Estimation. I estimate the choice model using data from the politics community given the relevance of this community to managers, policy makers, and users. I use individual-level comment decisions and

restrict the sample to users who comment on at least 20 articles in the periods I study, where a period consists of the 60 minutes following a Wayback Machine snapshot. Section 5.3.2 discusses the implications of this restriction on the interpretation of the findings.²²

I take this model to the data using a two-step procedure that simplifies the computation given the large number of periods, users, and posts. In the first step, I estimate the exposure parameters p_t and p_r . I estimate p_t by combining data on the number of users online at the start of each period, which are observed in the Wayback Machine snapshots, with public statements by the platform on the average session duration to estimate the number of users who log on to the platform during each period using Little's law (Little 1961). I then use public usage statistics again to estimate the number of active community members and calculate the share of active community members who log on in each period. Finally, I smooth estimates of p_t by taking the fitted values of a regression of the raw values of p_t on quarter and day of week fixed effects. The full details of this process are described in Section C.3 in the Online Appendix. I then estimate the remaining exposure parameters using an empirical analog of Equation (7) ($p_r = \exp\{-\sum_{r'=1}^r \tau_{r'} - 1\}$).

Given the estimates of p_t and p_r , I estimate the individual parameters β_i by maximizing the following log-likelihood function:

$$\mathcal{L}(\beta, \xi) = \sum_t \sum_i \sum_j d_{ijt} \log P_{ijt}(\beta_i, \xi_{jt}) + (1 - d_{ijt}) \log(1 - P_{ijt}(\beta_i, \xi_{jt})). \quad (8)$$

Finding the maximum likelihood estimate involves solving a high-dimensional optimization procedure due to the large number of individuals and posts. Therefore, I use the following iterative algorithm that solves the maximum likelihood problem subject to the constraint that observed comment shares equal predicted comment shares. First, I initialize a guess of δ_{jt} and, conditional on these fixed effects, estimate the individual-level preference parameters using maximum likelihood. I then invert observed engagement shares (Berry 1994) using the Berry et al. (1995) contraction mapping to find the values of δ_{jt} such that predicted market shares equal observed market shares.²³ I regress δ_{jt} on x_{jt} to recover β and the residual ξ_{jt} . I iterate between these two steps until convergence. Splitting the estimation algorithm into these two steps allows the maximum likelihood parameters to be estimated in parallel. Inference on the preference parameters uses cluster-robust standard errors.

The preference estimates of any individual user will contain sampling error. This implies that the distribution of preference estimates will be a convolution of the true distribution of preferences and sampling error,

leading the distribution of estimates to be overdispersed relative to the true distribution. To correct for this overdispersion, I shrink all preference estimates toward the grand-mean using the empirical Bayes procedure described in Section C.2 in the Online Appendix.

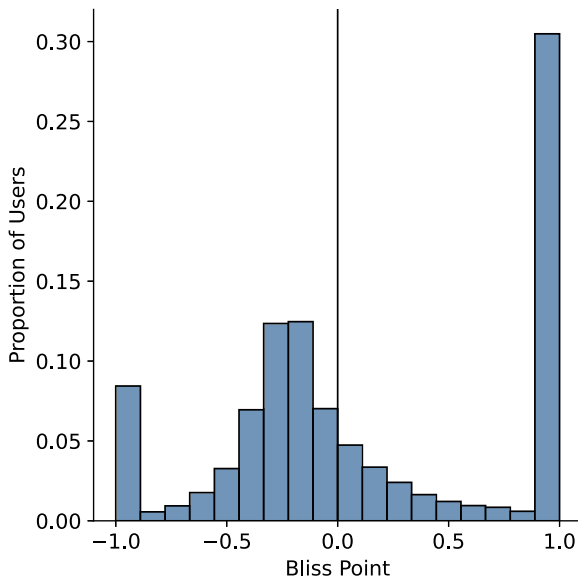
4.3. Model Estimates

I now discuss the model estimates and assess the fit of the model. Table C.5 in the Online Appendix summarizes the distribution of individual preference estimates ($\bar{\beta} + \beta_i$). It is helpful to summarize preferences for publisher slant through each user's bliss point, which is defined as the slant the user most prefers

$$b_i^* = \begin{cases} \text{sign}(\bar{\beta}_s + \beta_{is}) & \text{if } \bar{\beta}_{s^2} + \beta_{is^2} \geq 0 \\ \min\left\{1, \max\left\{-1, -\frac{\bar{\beta}_s + \beta_{is}}{2(\bar{\beta}_{s^2} + \beta_{is^2})}\right\}\right\} & \text{if } \bar{\beta}_{s^2} + \beta_{is^2} < 0, \end{cases} \quad (9)$$

where β_{is} (β_{is^2}) is user i 's taste parameter on post slant (slant squared). The marginal distribution of slant bliss points is shown in Figure 5. It is evident there is substantial preference heterogeneity in the politics community. More than half of users prefer less credible publishers, whereas the remaining users prefer more credible publishers. Regarding political slant, there is also substantial heterogeneity, with a large mass of users preferring outlets slightly left of center. There are also mass points at each political extreme, with nearly

Figure 5. (Color online) Distribution of Slant Bliss Points



Notes. This figure plots the marginal distribution of user-level slant bliss points. The bliss point is the slant score for which a user is most likely to comment, all else being equal. A bliss point of -1 implies the user is most likely to comment on left-leaning articles and a bliss point of 1 implies the user is most likely to comment on right-leaning articles.

10% of users preferring outlets that are strongly left leaning and roughly 30% of users preferring outlets that are strongly right leaning.

These revealed preference estimates reflect the types of content users tend to comment on and should not be interpreted as reflecting ideological agreement or a preference for this type of content. For example, it is plausible that users are more likely to comment on articles with which they disagree or find provocative, which may explain the large share of individuals who tend to comment on articles by low-credibility publishers. However, these revealed preferences are the signals passed to platforms and would be what the platform optimize for in their ranking algorithms. Therefore, these revealed preferences are the relevant primitive for the analysis of engagement patterns under counterfactual ranking algorithms in Section 5.

More broadly, the counterfactual results do not require these preference estimates to have a particular normative interpretation. The counterfactual exercise varies only the rank of articles on the feed, not article characteristics themselves. Therefore, the estimated mean utility of commenting on an article for a given user (δ_{ijt}) does not change across counterfactuals—only the probability of exposure does. This means that even if the credibility preference estimates partly reflect preferences over omitted features correlated with credibility, such as topic or writing style, the counterfactual predictions remain valid as long as the model accurately predicts commenting behavior, which I address next.

To assess the fit of the model, Table C.6 in the Online Appendix presents summary statistics of actual engagement and engagement predicted by the model. The distribution of actual engagement with engagement predicted by the model is also shown graphically in Figure C.24 in the Online Appendix. Figure C.24a demonstrates the high correlation between actual user engagement and predicted user engagement. Figure C.24b shows the correlation between actual and predicted user engagement by publisher credibility and the model again has a high correlation between actual and predicted engagement by group.

5. Counterfactual Ranking Algorithms

In this section, I use the model from Section 4 to analyze how counterfactual ranking algorithms impact user news diets. The goal of this analysis is to first understand how ranking content to maximize engagement impacts the credibility of news content with which users engage. Second, I evaluate the cost to the platform in terms of foregone engagement of credibility-aware ranking algorithms that balance total engagement and engagement with high-credibility content in the ranking objective function. I describe the mechanics of the

analysis of counterfactual ranking algorithms in Section 5.1, discuss the results in Section 5.2, and assess the robustness of the findings in Section 5.3.

5.1. Description of Counterfactual Rankings

I now describe the counterfactual ranking algorithms that I consider. In each period, the platform has an estimate of the probability that user i will comment on article j conditional on being exposed to that article ($\hat{P}_{ijt}$). The platform then ranks content to maximize the following objective functions.

Personalized engagement maximizing: A leading counterfactual considered is personalized engagement maximization. The personalized engagement-maximizing ranking solves

$$r_{it}^P = \arg \max_{r \in \mathcal{R}} \sum_{j=1}^J p(r_j, t) \hat{P}_{ijt}, \quad (10)$$

where $\mathcal{R}$ is the set of possible rankings and $r = \langle r_1, \dots, r_J \rangle \in \mathcal{R}$ is a vector of possible article ranks. It is straightforward to show that when $p(r, t)$ is decreasing in r , which is approximately true in this setting, the optimal ranking sorts articles in descending order of $\hat{P}_{ijt}$.²⁴ In words, this algorithm will rank content in decreasing order of the predicted engagement rate for each user.

Credibility-aware algorithm: Although short-term engagement is often used as a proxy for consumer welfare, a growing literature has emerged to study situations where these measures may differ. This disconnect can arise for rational economic agents (Spence and Owen 1977) and in models with behavioral biases, including agents with present bias (Kleinberg et al. 2023), dual self-models, (Kahneman 2011), and digital addiction (Allcott et al. 2022). Moreover, the platform may want to avoid promoting low-credibility publishers for brand safety purposes or to prevent potential regulatory actions. These factors could lead the platform to consider publisher credibility in the ranking objective function. Therefore, I consider a credibility-aware algorithm that maximizes an objective function that balances two competing objectives: total engagement and engagement with high-credibility publishers. Formally, the credibility-aware algorithm solves

$$r_{it}^O = \arg \max_{r \in \mathcal{R}} \sum_{j=1}^J p(r_j, t) ((1 - \lambda) + \lambda 1[c_{jt} \geq \underline{c}]) \hat{P}_{ijt}, \quad (11)$$

where λ reflects the weight on engagement above a minimum credibility threshold $\underline{c}$. Note that this nests the personalized engagement-maximizing algorithm when $\lambda = 0$ and a credibility-maximizing algorithm when $\lambda = 1$.²⁵ This algorithm ranks articles in descending order of the predicted objective function, $((1 - \lambda) + \lambda 1[c_{jt} \geq \underline{c}]) \hat{P}_{ijt}$, for each user. A nonpersonalized credibility-aware algorithm

sorts articles in descending order of the average index across users.

Nonpersonalized engagement maximizing: The nonpersonalized engagement-maximizing algorithm solves the following maximization problem:

$$r_t^N = \arg \max_{r \in \mathcal{R}} \sum_{j=1}^J p(r_j, t) E[\hat{P}_{ijt}], \quad (12)$$

which ranks articles in descending order of $E[\hat{P}_{ijt}]$ for all users. In words, this algorithm will rank content in descending order of the average predicted engagement rate.

Benchmarks: I compare the engagement patterns under the counterfactual algorithms described above to two benchmark algorithms: the ranking employed by the platform (Actual) and a random benchmark that randomly shuffles the articles shown on the page for each user (Random).

To form $\hat{P}_{ijt}$, I assume that the platform has high-quality estimates of user preferences given their access to rich user-level behavioral data and therefore assume the platform observes β_i in the counterfactuals. I assume the platform does not, however, observe latent article commentability (ξ_{jt}) and must estimate this using observable article features. I model the platform's estimates of ξ_{jt} as a supervised learning problem where the platform forms estimates of the true latent article commentability ($\hat{\xi}_{jt}$) based on article observables. I operationalize this using a random forest that predicts ξ_{jt} using observable post features including the stock of total and top-level comments, vote score, post age, publisher slant, and publisher credibility rating. This model performs well in the prediction task as demonstrated in Figure C.25 in the Online Appendix. With estimates of article commentability, observed post features, and observed user preferences, the platform can estimate engagement probabilities for each user and article conditional on exposure $\hat{P}_{ijt} = \frac{\exp \delta_{ijt}}{1 + \exp \delta_{ijt}}$, where $\delta_{ijt} = \mathbf{x}'_{jt}(\bar{\beta} + \beta_i) + \hat{\xi}_{jt}$. To calculate engagement under a counterfactual algorithm, I calculate engagement probabilities for each post and user by multiplying the exposure probability for the post under the counterfactual ranking with the true estimated engagement probability conditional on exposure.

In this study, I focus on the implications of different objective functions in algorithmic ranking conditional on the candidates available. That is, the counterfactuals considered here only rerank the top 25 posts in each period, which I treat as the set of candidate posts to be ranked.²⁶ This decision is relatively innocuous for analyzing the impact of optimizing for engagement, because latent post commentability is highly correlated with post rank in the data (Figure C.23 in the Online Appendix). Latent post commentability is an important

factor in optimizing for engagement, meaning posts that are not in the top 25 posts would be less likely to be ranked high on the feed even if they were included in the candidate posts. In settings with more diverse content, considering a larger set of candidate posts would likely be important.²⁷

5.2. Counterfactual Ranking Algorithm Results

Summaries of engagement patterns under the different counterfactual ranking algorithms are shown in Table 1. This table contains comparisons of the various counterfactual ranking algorithms relative to the platform’s actual ranking algorithm. Specifically, for three outcomes the table contains regression estimates of the following regression:

$$Y_{ia} = \gamma + \gamma_a + \varepsilon_{ia}, \tag{13}$$

where Y_{ia} is the outcome for user i under counterfactual ranking algorithm a , γ is the intercept, γ_a is a fixed effect for algorithm a , and ε_{ia} is an error term. The actual ranking algorithm is the excluded algorithm, so γ is the average outcome under the actual ranking algorithm and γ_a is the change in the average outcome from algorithm a relative to the actual ranking algorithm. Each observation is a user-algorithm pair. I now describe the impact the various ranking algorithms have on user news diets.

5.2.1. Impact on Engagement Quantity. The counterfactual analysis suggests that the algorithm employed by the platform, which prioritizes simplicity and transparency, is far from engagement maximizing. That

said, the actual algorithm does result in higher engagement relative to the random benchmark. As expected, optimizing for engagement leads to a substantial increase in engagement quantity.

Much of the benefit comes from ranking articles according to expected engagement without personalization, which is evidenced by the 50.2% increase in engagement under the nonpersonalized engagement-maximizing algorithm.

Personalizing user feeds increases engagement by 54.5% relative to the existing algorithm, providing a modest increase in engagement relative to the nonpersonalized engagement-maximizing algorithm. Although modest in size, this lift does demonstrate the platform has an incentive to personalize rankings to drive engagement. Given the scale of the platform, even modest increases in engagement in relative terms can be large in an absolute sense. The modest lift of personalizing relative to nonpersonalizing engagement maximization is likely due to the research setting. Focusing on the politics community limits the degree of heterogeneity in the content available and user preferences reducing the scope for personalization to increase engagement.²⁸ In addition, restricting to reranking the highest ranked 25 posts in each period and the form of preference heterogeneity that I allow for may contribute to the modest effect of personalization.²⁹

Finally, optimizing for engagement with high-credibility publishers also leads to a substantial increase in engagement relative to the actual algorithm employed (46.2%), but represents a substantial cost in terms of lost engagement relative to the engagement-maximizing algorithms.

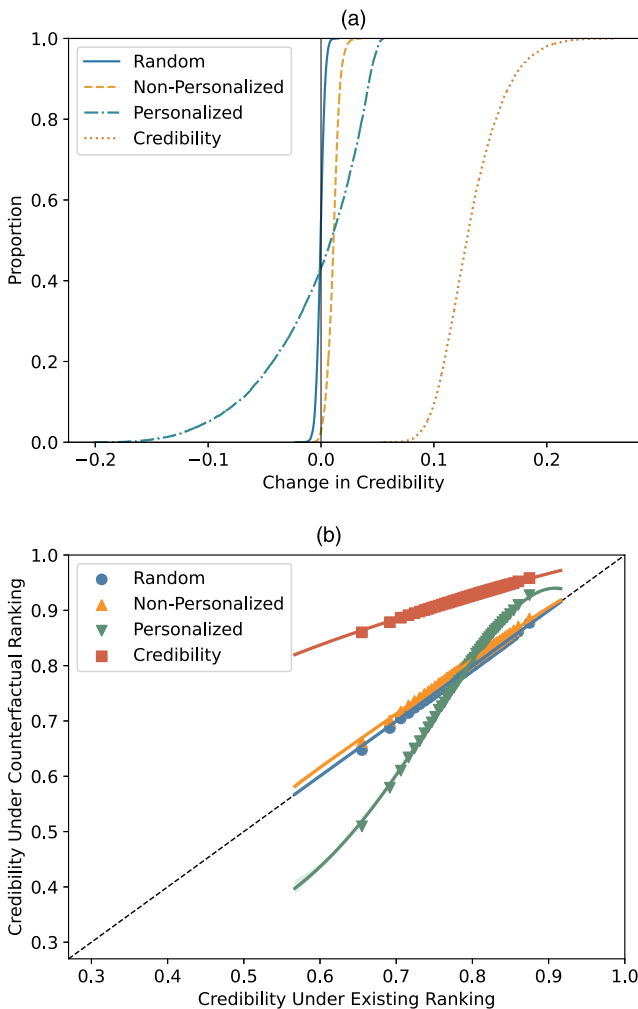
5.2.2. Impact on Engagement with Publishers by Credibility. Reddit’s algorithm does not materially impact the share of engagement with high-credibility publishers relative to a random ordering of posts, with both algorithms resulting in 79.0% of the average user’s engagement being with high-credibility publishers. Optimizing for engagement also does not lead to a substantial change for the average user, with an average high-credibility engagement share of 80.1% for the nonpersonalized engagement-maximizing algorithm and 78.6% for the personalized engagement-maximizing algorithm. The credibility-maximizing algorithm does lead to a substantial increase in the share of engagement with high-credibility publishers, with 92.3% of the average user’s engagement being with high-credibility publishers.

Focusing on average changes masks important heterogeneity. Figure 6(a) plots the empirical cumulative distribution function of the change in high-credibility shares relative to the existing algorithm. The nonpersonalized algorithm has only a modest impact on the quality of news diets as the share of engagement with

Table 1. Counterfactual Engagement Summaries

	Engagement	Distance to uniform	Credibility
Intercept	46.035 (0.304)	0.285 (0.001)	0.790 (0.000)
Random	−0.651 (0.009)	−0.006 (0.000)	−0.001 (0.000)
Nonpersonalized	23.115 (0.116)	−0.025 (0.000)	0.011 (0.000)
Personalized	25.099 (0.129)	0.036 (0.001)	−0.005 (0.000)
Credibility Max.	21.284 (0.112)	0.013 (0.001)	0.133 (0.000)
Observations	56,110	56,110	56,110
R ²	0.077	0.060	0.464

Notes. This table reports estimates of a panel regression of each counterfactual outcome on counterfactual algorithm dummy variables (Equation (13)). The intercept is the average quantity under the existing algorithm. (1) Engagement represents the total number of articles a user comments on. (2) Distance to uniform represents the first Wasserstein distance of engagement shares across publisher slant partitions from the uniform distribution. Recall distributions closer to uniform will have smaller distances, meaning they represent more diverse engagement. (3) Credibility represents the share of a users engagement with high-credibility publishers. Standard errors are clustered at the user level.

Figure 6. (Color online) Impact of Algorithm on Share of Engagement with High-Credibility Publishers

Notes. (a) Distribution of change in high-credibility share. (b) High-credibility share by baseline credibility.

high-credibility publishers slightly increases for nearly all users. The personalized engagement-maximizing algorithm, however, does have substantial impacts for many users despite the modest average effect. The majority of users experience a modest improvement in the quality of their news diets, because a slightly larger share of their engagement is with high-credibility publishers. However, 43.1% of users experience a deterioration in the quality of their news diets, with a subset of these users seeing the share of their engagement with high-credibility publishers falling by more than 10 percentage points. To better understand which users experience these declines, Figure 6(b) plots the relationship between news diet quality under the existing algorithm against news diet quality under the counterfactual algorithms. It is clear that users engaging with less credible publishers under Reddit's actual algorithm experience large declines in the quality of their news diets under

the personalized engagement-maximizing algorithm. This suggests the engagement-maximizing algorithm exacerbates differences in the quality of user news diets by promoting high-credibility publishers to the majority of users who typically engage with high-credibility publishers and promoting low-credibility publishers to users who have engaged with these publishers in the past. Moreover, the results are robust to the choice of threshold for high-credibility publishers (Section C.4 in the Online Appendix) and I find personalization exacerbates differences in the quality of user news diets even for very low thresholds for high-credibility publishers.

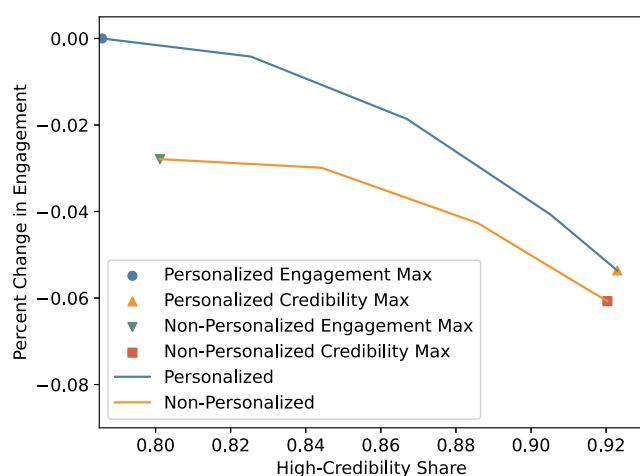
Turning to the credibility-maximizing algorithm, I find that optimizing for engagement with high-credibility publishers leads to substantial increases in the share of engagement with high-credibility publishers across all users. Importantly, however, Figure 6(b) shows that the users experiencing the largest increases are those who engage more with low-credibility publishers under Reddit's actual algorithm. This indicates that including publisher credibility in the objective function narrows the disparity between users with high- and low-quality news diets, a difference that was exacerbated when optimizing only for engagement.

5.2.3. Engagement-Credibility Tradeoff. The results thus far have compared engagement-maximizing algorithms with a credibility-maximizing algorithm. That said, platforms or society may balance these competing objectives in a more nuanced manner rather than preferring either extreme. I now describe the frontier of possible outcomes as λ , the weight placed on engagement with high-credibility publishers in the credibility-aware ranking algorithm, is varied.

Figure 7 plots this tradeoff along with points corresponding to the engagement-maximizing algorithm, credibility-maximizing algorithm, nonpersonalized engagement-maximizing algorithm, and nonpersonalized credibility-maximizing algorithm. As can be seen, moving to the credibility maximizing algorithm from the engagement-maximizing algorithm reduces engagement by 5.4% while at the same time reducing the average share of user engagement with low-credibility content by more than 60%. However, platforms can achieve over half of the increase in news diet quality from the credibility-maximizing algorithm for a 1.9% decrease in engagement. This change in engagement is slightly smaller in magnitude than the difference between the nonpersonalized engagement-maximizing algorithm and the personalized engagement-maximizing algorithm.

By contrast, the nonpersonalized algorithm does not meaningfully improve the quality of users' news diets, whereas the credibility-aware algorithm increases the average share of engagement with high-credibility

Figure 7. (Color online) Engagement-Quality Frontier



Notes. This figure plots the frontier of possible outcomes when varying λ in the credibility aware algorithm. The y-axis is the percent change in engagement relative to the personalized engagement-maximizing algorithm. The x-axis is the average share of engagement with high-credibility publishers. Points indicate outcomes under the counterfactual algorithms described in Section 5.1.

publishers by 7.5 percentage points for approximately the same total engagement.

The shape of this frontier is also important, because the gradient is relatively flat around the engagement-maximizing algorithms. This suggests that, for small decreases in engagement, the platform can drastically increase the share of engagement with high-quality publishers. However, this also means that small differences in preferences between the platform and society can lead to large discrepancies in outcomes along the credibility dimension, again highlighting the importance of aligning the ranking algorithm's objective function.

5.2.4. Additional Results. I now summarize additional impacts of the counterfactual ranking algorithms. First, I discuss the impact of the ranking algorithms on the quality of discussion that I operationalize through comment sentiment. Second, I discuss the heterogeneous impact on publisher market shares to understand how the various ranking algorithms impact publisher incentives. Finally, I discuss the impact on the diversity of engagement with publishers across the political spectrum.

First, I report findings on the impact optimizing for engagement has on discussion quality, as measured through the sentiment of comments submitted to the platform (See Online Appendix D for full analysis). Table D.11 in the Online Appendix demonstrates that both the nonpersonalized and personalized algorithms slightly elevate the share of negative-sentiment comments submitted by the average user relative to the existing algorithm. A negative-sentiment comment is significantly more likely to contain strongly negative

emotions such as anger and disgust and more likely to be classified as toxic (Pérez et al. 2021). Moreover, inspecting negative comments reveals they are often extremely vulgar and unlikely to meaningfully contribute to the discussion.

Although the effect on the sentiment of the average user is small, personalization increases the variance in sentiment leading to some users commenting more positively while others are shown content that makes them respond negatively (Figure D.37a in the Online Appendix). Figure D.37b plots the relationship between the change in the negative-sentiment share of users against user preferences for publisher credibility and I find that users who prefer less-credible publishers have a larger increase in their negative-sentiment share. The same is true of users who prefer left-leaning outlets, consistent with the sentiment preference estimates in Figure D.36 in the Online Appendix.

Next, I change the unit of analysis to the publisher and summarize how the counterfactual ranking algorithms impact different types of publishers. Figure C.27 in the Online Appendix plots the change in publisher market share by publisher slant (Figure C.27a) and publisher credibility (Figure C.27b).³⁰ Optimizing solely for engagement leads to a reallocation of market share from left-leaning publishers to right-leaning publishers and a slight increase in the market shares of low-credibility publishers. Optimizing for engagement with high-credibility publishers leads to a reallocation of engagement from politically slanted publishers to more neutral publishers and a reallocation from low- to high-credibility publishers.

Finally, I estimate the impact the counterfactual algorithms have on the diversity of engagement across the political spectrum. To do so, I discretize publisher slant into quintiles and calculate the first-order Wasserstein distance of engagement or promotion shares across these five bins of publisher slant relative to a uniform distribution. This distance metric is better suited to this setting relative to other common measures of diversity used in the literature, including the Herfindahl-Hirschman index and Shannon entropy, due to the ordered nature of slant partitions. For example, the Wasserstein distance between a user's engagement and the uniform distribution is larger (i.e., less diverse) for a user who engages only with publishers from a politically slanted partition versus a user only engaging with moderate publishers. The distance is minimized when users engage equally with publishers from all slant partitions and is largest when only engaging with publishers from a politically extreme partition.

The counterfactuals suggest that the random and nonpersonalized engagement-maximizing algorithms increase engagement diversity relative to the actual algorithm. That said, the personalized algorithm results in a decline in engagement diversity. This decline

occurs for the majority of users, with 64.5% of users experiencing a decline in their engagement diversity in the personalized engagement-maximizing counterfactual. Turning to the credibility-maximizing algorithm, I also find a decrease in the diversity of engagement as the average distance to uniform engagement shares rose by 4.6%.

5.3. Robustness and Limitations of Findings

I now highlight a number of limitations of the analysis and demonstrate the findings are robust to a number of alternative modeling choices. Section 5.3.1 demonstrates that the results presented above are robust to an alternative method of personalization that can be estimated on a much broader set of users to address concerns over selection and require a different set of assumptions relative to the model presented above. Section 5.3.2 addresses the concern over user selection directly, and Section 5.3.3 summarizes a version of the model that allows for users to reoptimize their beliefs over the type of articles users will encounter under the various counterfactual ranking algorithms. Finally, additional robustness exercises are presented in Section C.4 in the Online Appendix. This includes an analysis of the sensitivity of the results to the limited preference heterogeneity that I allow for in the model presented in the main text.

5.3.1. Robustness to Alternative Method of Personalization and External Validity. The results reported thus far are based on a microfounded model of user engagement decisions. This approach requires many assumptions, and here, I seek to show the results are not sensitive to the assumptions made. To do so, I summarize the findings of an analysis reported in Online Appendix E that analyzes the type of publishers that are promoted when personalizing the ranking algorithm to maximize engagement using a reduced form collaborative-filtering based recommender system (Hu et al. 2008). The recommender system then recommends publishers on which a user is most likely to comment in a period. I validate this recommender system by estimating heterogeneous treatment effects when the regression discontinuity experiments align with the recommender system's predictions. I find that the recommender system effectively predicts treatment effects, a result that suggests the model has learned important aspects of user preferences. I then study the types of content that gets promoted under this simple recommender system to understand the extent to which personalized engagement maximization impacts individual news diets.

This approach has two advantages over the discrete choice model and counterfactual analysis I study in Section 4 and Section 5. First, this model is trained using comment decisions from more than 500,000 users

and is evaluated on comment decisions of more than 180,000 users. This is a much larger sample than that used in the choice model approach, as I can use comment decisions on articles during periods not captured in Wayback Machine snapshots during the training process. Second, this approach relies on a different set of assumptions than the model of engagement decisions and can be validated by predicting treatment effects of which the model is predictive. I find consistent results across both approaches when comparable, which gives confidence that the findings of the choice model approach can be generalized to a broader set of users. That said, I emphasize the model-based approach as the main findings given it allows me to study in more detail the implications of various algorithm designs, including the credibility-aware ranking algorithms, on engagement rather than simply studying what content the algorithm would have recommended. In addition, the model-based approach takes seriously the impact rank has on engagement, which can bias the preferences learned by a model that does not account for this relationship (Chaney et al. 2018).

5.3.2. Selection. I now discuss how the restriction to users who comment on at least 20 articles in the periods I study impacts the interpretation of the results. Although the set of users who meet this criteria only represents 5% of users who submitted any comment during the periods I study, they represent 44% of comments submitted during these periods. Therefore, this sample represents an important share of engagement on the platform. Moreover, this sample includes users with only moderate activity. Commenting on 20 articles in my sample corresponds to averaging 1.6 comments per week.

However, it is worth asking how the analysis may change if the analysis included all users who ever comment on the platform. Although this is infeasible to implement in practice using the choice model, I can appeal to results presented in Section 5.3.1 that show the findings are consistent when using an alternative personalization algorithm that is evaluated on more than 180,000 users. In addition, I analyze how the results differ by user activity in my data. Figure C.29 in the Online Appendix plots the engagement-quality frontier estimated separately by the number of comments a user posted in the study period. The tradeoff between engagement and the credibility of publishers that users engage with is stable across the different groups. This provides confidence that the result is stable and would generalize to less engaged populations. In addition, I note that the study population represents a meaningful share of engagement in the politics community and is an important population for the platform given the platform's focus on user-generated content.

5.3.3. Robustness to Endogenous Search. The analysis above holds fixed user attention across the feed. In equilibrium one might expect users to reallocate their attention in response to the ranking algorithms. Section C.4.4 in the Online Appendix estimates a version of the model where I decompose the treatment effects of rank on engagement into two quantities: search costs and the expected utility of viewing an article in a given position. This is consistent with recent work studying the impact of ranking algorithms on consumer behavior (Fong et al. 2024, Kaye 2024). I find that search costs are the primary reason that rank impacts engagement and therefore the engagement patterns I estimate under counterfactual ranking algorithms are similar when I allow for consumers to reoptimize their attention across the feed.

5.3.4. Additional Limitations. Although the above robustness exercises address some limitations of the primary model, additional limitations remain. In particular, this is a partial equilibrium analysis that holds several dimensions fixed. Although this model is able to capture some endogenous response to the algorithm, I do not allow substitution away from the platform or substitution to other ranking algorithms.³¹ In addition, I hold fixed the distribution of content that is supplied to the platform, voting patterns that may influence engagement and ranking, and broader dynamics of attention and discussion. Analyzing how these dimensions respond to ranking algorithms is a promising area for future work and likely requires additional variation in the ranking algorithm beyond what is captured in my data.

6. Discussion and Conclusion

In this study, I evaluate the impact of optimizing for engagement in social media news feed algorithms on user news diets. To address this question, I exploit a regression discontinuity revealed in the platform's code to identify the causal effect of rank on engagement and use these causal estimates to identify a model of user engagement. Using this model, I estimate engagement patterns under counterfactual ranking algorithms including personalized and nonpersonalized engagement-maximizing and a credibility-aware algorithm that explicitly trades off total engagement and engagement with high-credibility publishers.

The counterfactual analysis demonstrates that social media platforms have a strong incentive to optimize their ranking algorithms for engagement. Optimizing for engagement leads to a dramatic increase in the quantity of engagement, and much of this results from promoting posts with which all users are likely to engage. The marginal benefit of personalizing feeds is modest in terms of engagement quantity but has substantial

impacts on the credibility and diversity of publishers with which users engage. In particular, personalized engagement maximization exacerbates differences in the quality of user news diets. That is, the personalized engagement-maximizing ranking increases inequality in the share of high-credibility engagement between users engaging with lower-credibility publishers and those engaging with higher-credibility publishers under the existing algorithm.

Advertiser concerns about brand safety give platform managers a direct motive to promote credible publishers. Many advertisers seek to avoid advertising on platforms that promote content that is inconsistent with their values or that would create backlash from their consumers. For example, the #StopHate-ForProfit movement led more than 1,000 large advertisers to halt or reduce advertising on Facebook to pressure the platform to expand its efforts to combat hate speech and misinformation (Hsu and Friedman 2020). There is also evidence suggesting that firms advertising on platforms alongside misinformation can experience customer backlash (Ahmad et al. 2024).

The credibility-aware algorithm demonstrates one method managers can use to improve the credibility of news content that is promoted on their platforms. The gradient of the engagement-credibility frontier indicates that moving away from the engagement-maximizing algorithm and toward the credibility-maximizing algorithm incurs a relatively small cost in terms of lost engagement. However, this also implies that small differences in the preferences of the platform and society can generate large changes in the amount of engagement with low-credibility publishers despite reasonably small changes to total engagement.

The results are positive in nature, and I refrain from making normative claims about the weight publishers should place on publisher credibility. Eliciting the preferences of relevant stakeholders—including platform users, managers, creators, and broader society—over different algorithms is a promising avenue of future research. As one example, it is possible that placing a positive weight on publisher credibility stifles the entry of new publishers who do not have an established reputation. In addition, this analysis should not be taken to suggest that quantity of engagement and the credibility of content with which users engage are the only two dimensions entering the platforms or society's objective function. For example, platforms or society may have preferences over the nature of comments users submit. If this is the case, that I find engagement-maximizing algorithms increase the variance of users' negative engagement share would likely be a relevant finding for managers and policymakers. Rather than arguing quantity and credibility are the only two dimensions to care about, I view these two dimensions as likely

important to both platforms and society, but do not claim they are exhaustive.

In addition, these results are also relevant for managers of publishers and the incentives they face when advertising revenue on traffic originating from social media referrals comprises an important component of their income. I find that personalized engagement maximization benefits publishers with a strong conservative slant in addition to those who are less-credible. This introduces an incentive for publishers to change their coverage to match the increased demand for politically slanted and low-credibility journalism.

Finally, these results have implications for regulating digital platforms. A growing regulatory trend is to require or incentivize platforms to allow users to opt-out of personalized recommendations or feeds. Examples include the European Union's Digital Services Act or proposed legislation (such as the Filter Bubble Transparency Act, Justice Against Malicious Algorithms Act, and the Protecting Americans from Dangerous Algorithms Act) in the United States. The findings presented here suggest the emphasis on allowing users to opt out of personalization may be misguided. Rather, as the results show, personalization can have substantial benefits when the objective function aligns with society's preferences. Recall that for approximately the same level of engagement as the nonpersonalized engagement-maximizing algorithm, the credibility-aware algorithm can increase the share of the average user's engagement with high-credibility publishers by 7.5 percentage points. A solution that takes advantage of the benefits of personalization, while protecting individual autonomy, could be to allow users to adjust the weights on different components within the ranking algorithm objective function, including the weight placed on publisher credibility. What weights users would choose and the results of such a design remain open questions and merit future work. Alternatively, regulators could incentivize platforms to align their ranking objective function with the preferences of society to take advantage of personalized ranking algorithms' substantial benefits while mitigating potential negative effects, although implementing such a policy would face substantial legal and ethical challenges.

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seminar participants for helpful conversations, comments, and suggestions.

Endnotes

¹ The politics community is consistently ranked as one of the most active communities on the platform.

² Despite the algorithm changing since 2017, the results from this analysis remain important. In particular, much of the paper considers the tradeoff between engagement and the credibility of content with which users engage. This comparison does not involve Reddit's actual algorithm, and to the extent that platforms attempt optimize rankings for engagement, the findings become increasingly important.

³ This identification strategy is most closely related to Narayanan and Kalyanam (2015), where data on the AdRank scores in Google auctions are used to estimate the position effects on Google advertisements, although to my knowledge, this is the first application of such a strategy to a social media setting.

⁴ See <https://www.redditinc.com/>.

⁵ On the platform, this default algorithm is called the *hot* algorithm. I refer to the hot algorithm as the actual or baseline ranking algorithm throughout.

⁶ In 2018, after the period studied, Reddit changed the default algorithm used by the Home feed to the Best algorithm, as described in https://www.reddit.com/r/changelog/comments/7spgg0/best_is_the_new_hotness/.

⁷ In a supplemental survey, I estimate that approximately 40% of users engage with Reddit through the Home feed while the remaining users navigate to the community page (Figure A.6).

⁸ Occasionally the Wayback Machine will capture multiple snapshots in rapid succession. I therefore restrict to snapshots that are separated by at least one hour to avoid such duplicates.

⁹ Posts pinned by moderators are shown in green and are typically threads created to discuss the major events of the day. Importantly, these are not algorithmically ranked, and I condition on these posts remaining in their position on the feed. That is, I only consider counterfactuals where the organic post positions change.

¹⁰ The threshold of 0.65 is chosen as it is the median Lin et al. (2022) credibility rating within the Medium credibility category of Media Bias Fact Check, a professional rating organization.

¹¹ Optimal bandwidth selection procedures are not readily available for nonlinear regression models. Therefore, I use the mean squared error minimizing bandwidth for a linear regression discontinuity with log of one plus the number of comments received immediately following a snapshot and show robustness to alternative bandwidths.

¹² I refer to quasi-Poisson regression as Poisson regression throughout for notational brevity.

¹³ See <https://www.reddit.com/wiki/faq>.

¹⁴ See <https://web.archive.org/web/20170121192832> and <https://redditblog.com/2017/1/17/caching-at-reddit/>.

¹⁵ This pattern appears in other settings where position effects have been credibly estimated in the past (Figure B.12), although the concentration of overestimation at the top position in the feed is more stark in this setting.

¹⁶ Section 5.3 endogenizes the search process by allowing a user to decide whether to consider an article in position r by comparing the expected benefit of viewing articles in position r to a search cost of doing so.

¹⁷ A key simplification is the stylized process by which users form consideration sets. A more flexible model that allows users to consider a subset of posts and comment on their most preferred post

introduces substantial computational challenges as the number of potential consideration sets grows combinatorially. These models would likely yield similar results given users are highly unlikely to comment on more than one post in either the data or in the counterfactual simulations. Therefore, given the substantial computational benefits of assuming independence between comment decisions, I assume that users make engagement decisions on posts independently.

¹⁸ The existing literature has studied many motivations and incentives for why users contribute user-generated content, including social norms (Chen et al. 2010, Burtch et al. 2018), and status or rewards (Goes et al. 2016, Gallus 2017, Burtch et al. 2021). Unpacking individuals' underlying motivations for commenting is beyond the scope of this analysis; rather, I estimate the utility preferences and use them as an input for analyzing engagement patterns under counterfactual ranking algorithms.

¹⁹ The latent commentability term ξ_{jt} is often referred to as latent quality in the literature estimating demand systems (Berry 1994). To avoid confusion with publisher credibility, I refer to ξ_{jt} as latent commentability, where this captures a vertical component making all users more likely to comment on the article.

²⁰ This assumption is motivated by the findings of Ursu (2018) that demonstrates empirically that rankings impact search probabilities but, conditional on search, do not affect purchase probabilities in an online travel platform. This is also consistent with recent work modeling personalized rankings in e-commerce (Donnelly et al. 2023).

²¹ The results are robust to other choices of p_1 as shown in the Online Appendix Section C.4. This could arise if users who are online are actively viewing content on other subreddits and thus not exposed to the content from the politics subreddit.

²² A user commenting on 20 articles in my sample corresponds to commenting 1.6 times per week on average.

²³ There are instances in the data where a post receives zero comments. I assume that ξ_{jt} is bounded below such that the minimum predicted market share is equal to 0.01% in these situations.

²⁴ To show this, assume for contradiction there exists an optimal ranking with two posts j and j' such that $r_j < r_{j'}$ and $\hat{P}_{ijt} < \hat{P}_{ij't}$. Note that the objective under this ranking is less than the objective if the positions of the two posts are swapped $(\hat{P}_{ijt} - \hat{P}_{ij't})(p(r_j, t) - p(r_{j'}, t)) > 0$ because $p(\cdot)$ is decreasing in r . Therefore, this ranking is not optimal, thus providing a contradiction.

²⁵ When $\lambda = 1$, high-credibility articles are ranked above low-credibility articles, with ordering within each group determined by predicted engagement.

²⁶ I restrict to algorithms that re-rank the top 25 posts in each period because I only observe the position of the top 25 articles in each Wayback Machine snapshot. I therefore am unable to estimate treatment effects of rank on engagement and article-period fixed effects (ξ_{jt}) for articles ranked lower on the page, which are key inputs for calculating engagement under counterfactual ranking algorithms.

²⁷ This assumption does preclude analyzing simple proposed algorithms such as reverse chronological, because the restricted set of candidate posts excludes the high volume of low-quality posts that are often promoted under a reverse-chronological ranking.

²⁸ This is consistent with the existing literature. Peukert et al. (2023) find that personalized recommendations on a news website increase clicks by 2.5% relative to nonpersonalized expert recommendations. In contrast, Holtz et al. (2020) finds that a personalized podcast recommender increases engagement by 28.9%: a setting with a substantially larger share of preference heterogeneity.

²⁹ Online Appendix C.4.1 considers a model with richer preference heterogeneity and finds that the personalized engagement-

maximizing algorithm only increases engagement by a modest 5.5% relative to the nonpersonalized engagement-maximizing algorithm.

³⁰ Here, publisher market share is defined as a publisher's share of total engagement in the counterfactuals. This differs from how publisher market share would traditionally be defined, and one should think of market share in this context as the share of traffic from the platform.

³¹ In a supplementary survey of Reddit users, I estimate that a large majority of users report they would not substitute away from the platform in response to a credibility-aware ranking algorithm (Figure A.7).

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